Node Structure Visualization

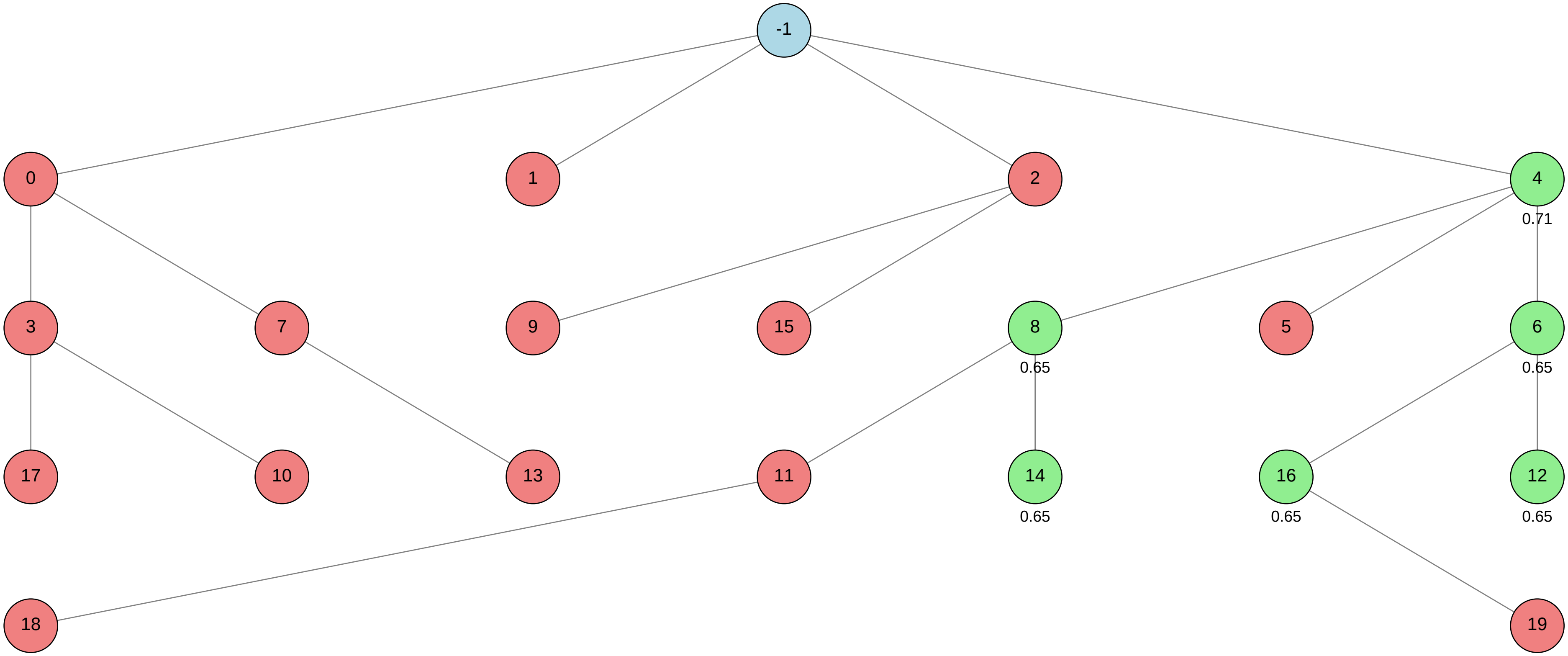
Click on any node in the tree to jump to its detailed information.

Total Nodes: 21 | Best Validation Score: 0.7085 (Node 4)

Node Tree:

Table of Contents:

Node ID	Stage	Status	Tool
-1	root		
0	evolve	✗	autogluon.multimodal
1	evolve	✗	machine learning
2	evolve	✗	autogluon.tabular
3	debug	✗	autogluon.multimodal
4	debug	✓	machine learning
5	evolve	✗	machine learning
6	evolve	✓	machine learning
7	debug	✗	autogluon.multimodal
8	debug	✓	machine learning
9	debug	✗	autogluon.tabular
10	debug	✗	autogluon.multimodal
11	evolve	✗	machine learning
12	evolve	✓	machine learning
13	debug	✗	autogluon.multimodal
14	evolve	✓	machine learning
15	debug	✗	autogluon.tabular
16	evolve	✓	machine learning
17	debug	✗	autogluon.multimodal
18	debug	✗	machine learning
19	evolve	✗	machine learning



Status	Color
Success	Green
Failure	Red
Neutral	Blue

Node Details:

Node -1 (root)

Property	Value
uct_value	-0.2500
ctime	2026-05-22 16:55:42
debug_attempts	0
depth	0
execution_time	0.0
expected_child_count	0
failure_visits	14
id	-1
is_debug_successful	✗
is_leaf	✗
is_successful	✗
is_terminal	✗
num_children	4
prev_tutorial_prompt	None
stage	root
time_step	-1
tool_used	
tools_available	[]
unvalidated_visits	0
validated_reward	3.9370000000000007
validated_visits	6
validation_score	None
visits	20
parent_id	None
child_ids	[0, 1, 2, 4]

Node 0 (evolve)

Property	Value
uct_value	0.6658
ctime	2026-05-22 16:56:38
debug_attempts	2
depth	1
execution_time	0.0
expected_child_count	0
exploitation	-0.3
exploration	1.06439799203612
failure_penalty	-0.3
failure_visits	6
id	0
is_debug_successful	✗
is_leaf	✗
is_successful	✗
is_terminal	✓
normalized_failure_visit	3
num_children	2
prev_tutorial_prompt	None
stage	evolve
time_step	0
tool_used	autogluon.multimodal
tools_available	['autogluon.multimodal', 'machine learning', 'autogluon.tabular']
unvalidated_visits	0
validated_contribution	0.0
validated_reward	0.0
validated_visits	0
validation_score	None

visits	6
parent_id	-1
child_ids	[3, 7]

Error Message:

stderr:
==> WARNING: A newer version of conda exists. <==
current version: 25.11.0
latest version: 26.5.0
Please update conda by running
$ conda update -n base -c conda-forge conda
Using Python 3.11.15 environment at: 48-addition_2/node_0/conda_env
Resolved 235 packages in 1.26s
Uninstalled 1 package in 2.11s
Installed 233 packages in 1m 49s
+ absl-py==2.4.0
+ accelerate==1.13.0
+ adagio==0.2.6
+ aiohappyeyeballs==2.6.2
+ aiohttp==3.13.5
+ aiohttp-cors==0.8.1
+ aiosignal==1.4.0
+ alembic==1.18.4
+ annotated-doc==0.0.4
+ annotated-types==0.7.0
+ antlr4-python3-runtime==4.9.3
+ attrs==26.1.0
+ autogluon==1.5.1b20260522
+ autogluon-common==1.5.1b20260522
+ autogluon-core==1.5.1b20260522
+ autogluon-features==1.5.1b20260522
+ autogluon-multimodal==1.5.1b20260522
+ autogluon-tabular==1.5.1b20260522
+ autogluon-timeseries==1.5.1b20260522
+ beartype==0.22.9
+ beautifulsoup4==4.14.3
+ blis==1.3.3
+ boto3==1.43.13
+ botocore==1.43.13
+ catalogue==2.0.10
+ catboost==1.2.10
+ certifi==2026.5.20
+ cffi==2.0.0

+ chardet==7.4.3
+ charset-normalizer==3.4.7
+ chronos-forecasting==2.2.2
+ click==8.4.1
+ cloudpathlib==0.24.0
+ cloudpickle==3.1.2
+ colorama==0.4.6
+ colorful==0.6.0a1
+ colorlog==6.10.1
+ confection==1.3.3
+ contourpy==1.3.3
+ coreforecast==0.0.16
+ cryptography==48.0.0
+ cycler==0.12.1
+ cymem==2.0.13
+ datasets==4.0.0
+ defusedxml==0.7.1
+ dill==0.3.8
+ distlib==0.4.0
+ einops==0.8.2
+ einx==0.4.3
+ evaluate==0.4.6
+ fastai==2.8.5
+ fastcore==1.8.18
+ fastdownload==0.0.7
+ fastprogress==1.0.5
+ fasttransform==0.0.2
+ filelock==3.29.0
+ fonttools==4.63.0
+ frozendict==2.4.7
+ frozenlist==1.8.0
+ fsspec==2025.3.0
+ fugue==0.9.7
+ future==1.0.0
+ gdown==6.0.0
+ gluonts==0.16.2
+ google-api-core==2.30.3
+ google-auth==2.53.0
+ googleapis-common-protos==1.75.0
+ graphviz==0.21
+ grpcio==1.81.0rc1
+ hf-xet==1.5.0
+ httpx==1.0.dev3
+ huggingface-hub==0.36.2
+ hyperopt==0.2.7

+ idna==3.16
... [truncated, 6684 chars total]

Error Analysis:

ERROR_SUMMARY: The error occurs because the `MultiModalPredictor.fit()` method from AutoGluon does not accept a `val_data` keyword argument. Instead, validation data should be provided using the `tuning_data` parameter, or by relying on AutoGluon's internal validation split if `tuning_data` is not specified.

SUGGESTED_FIX:
Remove the `val_data=val_data` argument from the `predictor.fit()` call and replace it with `tuning_data=val_data`. Alternatively, omit the validation split argument entirely to let AutoGluon handle validation automatically, or consult the [AutoGluon MultiModal documentation](https://auto.gluon.ai/stable/tutorials/multimodal/index.html) for the latest supported parameters.

Node 1 (evolve)

Property	Value
uct_value	2.4474
ctime	2026-05-22 17:06:34
debug_attempts	0
depth	1
execution_time	0.0
expected_child_count	0
exploitation	0.0
exploration	1.2486433288776608
failure_penalty	-0.0
failure_visits	1
id	1
is_debug_successful	✗
is_leaf	✓
is_successful	✗
is_terminal	✓
normalized_failure_visit	0
num_children	0
prev_tutorial_prompt	None
stage	evolve
time_step	1
tool_used	machine learning
tools_available	['autogluon.multimodal', 'machine learning', 'autogluon.tabular']
unvalidated_visits	0
validated_contribution	0.0
validated_reward	0.0
validated_visits	0
validation_score	None

visits	1
parent_id	-1
child_ids	[]

Error Message:

stderr:
==> WARNING: A newer version of conda exists. <==
current version: 25.11.0
latest version: 26.5.0
Please update conda by running
$ conda update -n base -c conda-forge conda
warning: Requirements file `/home/anri21/.conda/envs/mlzero_env/lib/python3.11/site-packages/autogluon/assistant/tools_registry/machine learning/requirements.txt` does not contain any dependencies
Using Python 3.11.15 environment at: 48-addition_2/node_1/conda_env
Resolved 79 packages in 1.09s
Uninstalled 1 package in 2.06s
Installed 78 packages in 40.95s
+ aiohappyeyeballs==2.6.2
+ aiohttp==3.13.5
+ aiosignal==1.4.0
+ annotated-doc==0.0.4
+ annotated-types==0.7.0
+ anyio==4.13.0
+ attrs==26.1.0
+ certifi==2026.5.20
+ chardet==7.4.3
+ charset-normalizer==3.4.7
+ click==8.4.1
+ cuda-bindings==13.2.0
+ cuda-pathfinder==1.5.4
+ cuda-toolkit==13.0.2
+ datasets==4.8.5
+ dill==0.4.1
+ filelock==3.29.0
+ frozenlist==1.8.0
+ fsspec==2026.2.0
+ h11==0.16.0
+ hf-xet==1.5.0
+ httpcore==1.0.9
+ httpx==0.28.1
+ huggingface-hub==1.16.1
+ idna==3.16
+ jinja2==3.1.6

+ markdown-it-py==4.2.0
+ markupsafe==3.0.3
+ mdurl==0.1.2
+ mpmath==1.3.0
+ multidict==6.7.1
+ multiprocess==0.70.19
+ networkx==3.6.1
+ numpy==2.4.6
+ nvidia-cublas==13.1.1.3
+ nvidia-cuda-cupti==13.0.85
+ nvidia-cuda-nvrtc==13.0.88
+ nvidia-cuda-runtime==13.0.96
+ nvidia-cudnn-cu13==9.20.0.48
+ nvidia-cufft==12.0.0.61
+ nvidia-cufile==1.15.1.6
+ nvidia-curand==10.4.0.35
+ nvidia-cusolver==12.0.4.66
+ nvidia-cusparse==12.6.3.3
+ nvidia-cusparselt-cu13==0.8.1
+ nvidia-nccl-cu13==2.29.7
+ nvidia-nvjitlink==13.0.88
+ nvidia-nvshmem-cu13==3.4.5
+ nvidia-nvtx==13.0.85
+ opencv-python==4.13.0.92
+ pandas==3.0.3
+ pillow==12.2.0
+ propcache==0.5.2
+ pyarrow==24.0.0
+ pydantic==2.14.0a1
+ pydantic-core==2.47.0
+ pygments==2.20.0
+ python-calamine==0.6.2
+ python-dateutil==2.9.0.post0
+ pytz==2026.2
+ pyyaml==6.0.3
+ requests==2.34.2
+ rich==15.0.0
- setuptools==82.0.1
+ s... [truncated, 29939 chars total]

Error Analysis:

ERROR_SUMMARY: The script raises a KeyError when attempting to extract the columns ['skin_tone', 'alternative_skin_tone'] from the test DataFrame in the extract_features_df function. This occurs because test.csv does not contain these columns, but the code assumes their presence for both train and test data, leading to failure during feature extraction for the test set.

SUGGESTED_FIX:
Modify the extract_features_df function to check if the DataFrame contains the 'skin_tone' and 'alternative_skin_tone' columns before attempting to extract them. If these columns are missing (as in the test set), fill in default values (e.g., -1) or omit them, ensuring the feature dimensions match between train and test. This will allow the script to process test data without error.

Node 2 (evolve)

Property	Value
uct_value	1.2463
ctime	2026-05-22 17:12:37
debug_attempts	2
depth	1
execution_time	0.0
expected_child_count	0
exploitation	-0.16666666666666666
exploration	0.700422462826868
failure_penalty	-0.16666666666666666
failure_visits	3
id	2
is_debug_successful	✗
is_leaf	✗
is_successful	✗
is_terminal	✗
normalized_failure_visit	1
num_children	2
prev_tutorial_prompt	None
stage	evolve
time_step	2
tool_used	autogluon.tabular
tools_available	['autogluon.multimodal', 'machine learning', 'autogluon.tabular']
unvalidated_visits	0
validated_contribution	0.0
validated_reward	0.0
validated_visits	0
validation_score	None

visits	3
parent_id	-1
child_ids	[9, 15]

Error Message:

stderr:
==> WARNING: A newer version of conda exists. <==
current version: 25.11.0
latest version: 26.5.0
Please update conda by running
$ conda update -n base -c conda-forge conda
Using Python 3.11.15 environment at: 48-addition_2/node_2/conda_env
Resolved 239 packages in 3.17s
Uninstalled 1 package in 1.97s
Installed 237 packages in 1m 47s
+ absl-py==2.4.0
+ accelerate==1.13.0
+ adagio==0.2.6
+ aiohappyeyeballs==2.6.2
+ aiohttp==3.13.5
+ aiohttp-cors==0.8.1
+ aiosignal==1.4.0
+ alembic==1.18.4
+ annotated-doc==0.0.4
+ annotated-types==0.7.0
+ antlr4-python3-runtime==4.9.3
+ attrs==26.1.0
+ autogluon==1.5.1b20260522
+ autogluon-common==1.5.1b20260522
+ autogluon-core==1.5.1b20260522
+ autogluon-features==1.5.1b20260522
+ autogluon-multimodal==1.5.1b20260522
+ autogluon-tabular==1.5.1b20260522
+ autogluon-timeseries==1.5.1b20260522
+ beartype==0.22.9
+ beautifulsoup4==4.14.3
+ blis==1.3.3
+ boto3==1.43.13
+ botocore==1.43.13
+ catalogue==2.0.10
+ catboost==1.2.10
+ certifi==2026.5.20
+ cffi==2.0.0

+ chardet==7.4.3
+ charset-normalizer==3.4.7
+ chronos-forecasting==2.2.2
+ click==8.4.1
+ cloudpathlib==0.24.0
+ cloudpickle==3.1.2
+ colorama==0.4.6
+ colorful==0.6.0a1
+ colorlog==6.10.1
+ confection==1.3.3
+ contourpy==1.3.3
+ coreforecast==0.0.16
+ cryptography==48.0.0
+ cycler==0.12.1
+ cymem==2.0.13
+ daal==2025.9.0
+ datasets==4.0.0
+ defusedxml==0.7.1
+ dill==0.3.8
+ distlib==0.4.0
+ einops==0.8.2
+ einx==0.4.3
+ evaluate==0.4.6
+ fastai==2.8.5
+ fastcore==1.8.18
+ fastdownload==0.0.7
+ fastprogress==1.0.5
+ fasttransform==0.0.2
+ filelock==3.29.0
+ fonttools==4.63.0
+ frozendict==2.4.7
+ frozenlist==1.8.0
+ fsspec==2025.3.0
+ fugue==0.9.7
+ future==1.0.0
+ gdown==6.0.0
+ gluonts==0.16.2
+ google-api-core==2.30.3
+ google-auth==2.53.0
+ googleapis-common-protos==1.75.0
+ graphviz==0.21
+ grpcio==1.81.0rc1
+ hf-xet==1.5.0
+ httpx==1.0.dev3
+ huggingface-hub==0.36.2

+ hyperopt==0.... [truncated, 83394 chars total]

Error Analysis:

ERROR_SUMMARY: AutoGluon failed to train any models because the Ray backend attempted to create a UNIX domain socket with a path longer than the system limit (107 bytes), resulting in an OSError: "validate_socket_filename failed: AF_UNIX path length cannot exceed 107 bytes". This is due to excessively long directory paths in the temporary scratch space, which Ray (used internally by AutoGluon for parallelism) cannot handle for its socket files.

SUGGESTED_FIX:
Shorten the directory paths used for temporary files and output, especially the scratch and working directories (e.g., /sc-scratch/sc-scratch-ikim-guidlight/...). Move your project to a location with a much shorter absolute path (ideally under /tmp or a short-named directory), and set the TMPDIR, TEMP, and TMP environment variables to this shorter path before running your script to ensure Ray and AutoGluon use it for all temporary files. Then rerun your training script.

Node 3 (debug)

Property	Value
uct_value	0.9261
ctime	2026-05-22 17:20:27
debug_attempts	3
depth	2
execution_time	0.0
expected_child_count	0
exploitation	0.0
exploration	1.2684446626418298
failure_penalty	-0.0
failure_visits	3
id	3
is_debug_successful	✗
is_leaf	✗
is_successful	✗
is_terminal	✓
normalized_failure_visit	0
num_children	2
prev_tutorial_prompt	### Condensed: ```python # Condensed: ```python Summary: This tutorial demonstrates image classification using AutoGluon MultiModal, covering implementatio
stage	debug
time_step	3
tool_used	autogluon.multimodal
tools_available	['autogluon.multimodal', 'machine learning', 'autogluon.tabular']
unvalidated_visits	0
validated_contribution	0.0
validated_reward	0.0

validated_visits	0
validation_score	None
visits	3
parent_id	0
child_ids	[17, 10]

Error Message:

stderr:
==> WARNING: A newer version of conda exists. <==
current version: 25.11.0
latest version: 26.5.0
Please update conda by running
$ conda update -n base -c conda-forge conda
Using Python 3.11.15 environment at: 48-addition_2/node_3/conda_env
Resolved 235 packages in 1.46s
Uninstalled 1 package in 2.20s
Installed 233 packages in 1m 44s
+ absl-py==2.4.0
+ accelerate==1.13.0
+ adagio==0.2.6
+ aiohappyeyeballs==2.6.2
+ aiohttp==3.13.5
+ aiohttp-cors==0.8.1
+ aiosignal==1.4.0
+ alembic==1.18.4
+ annotated-doc==0.0.4
+ annotated-types==0.7.0
+ antlr4-python3-runtime==4.9.3
+ attrs==26.1.0
+ autogluon==1.5.1b20260522
+ autogluon-common==1.5.1b20260522
+ autogluon-core==1.5.1b20260522
+ autogluon-features==1.5.1b20260522
+ autogluon-multimodal==1.5.1b20260522
+ autogluon-tabular==1.5.1b20260522
+ autogluon-timeseries==1.5.1b20260522
+ beartype==0.22.9
+ beautifulsoup4==4.14.3
+ blis==1.3.3
+ boto3==1.43.13
+ botocore==1.43.13
+ catalogue==2.0.10

+ catboost==1.2.10
+ certifi==2026.5.20
+ cffi==2.0.0
+ chardet==7.4.3
+ charset-normalizer==3.4.7
+ chronos-forecasting==2.2.2
+ click==8.4.1
+ cloudpathlib==0.24.0
+ cloudpickle==3.1.2
+ colorama==0.4.6
+ colorful==0.6.0a1
+ colorlog==6.10.1
+ confection==1.3.3
+ contourpy==1.3.3
+ coreforecast==0.0.16
+ cryptography==48.0.0
+ cycler==0.12.1
+ cymem==2.0.13
+ datasets==4.0.0
+ defusedxml==0.7.1
+ dill==0.3.8
+ distlib==0.4.0
+ einops==0.8.2
+ einx==0.4.3
+ evaluate==0.4.6
+ fastai==2.8.5
+ fastcore==1.8.18
+ fastdownload==0.0.7
+ fastprogress==1.0.5
+ fasttransform==0.0.2
+ filelock==3.29.0
+ fonttools==4.63.0
+ frozendict==2.4.7
+ frozenlist==1.8.0
+ fsspec==2025.3.0
+ fugue==0.9.7
+ future==1.0.0
+ gdown==6.0.0
+ gluonts==0.16.2
+ google-api-core==2.30.3
+ google-auth==2.53.0
+ googleapis-common-protos==1.75.0
+ graphviz==0.21
+ grpcio==1.81.0rc1
+ hf-xet==1.5.0

+ httpx==1.0.dev3
+ huggingface-hub==0.36.2
+ hyperopt==0.2.7
+ idna==3.16
... [truncated, 6628 chars total]

Error Analysis:

ERROR_SUMMARY: The error occurs because the `fit()` method of `MultiModalPredictor` does not accept an `eval_metric` keyword argument. Passing this unsupported argument results in a `TypeError`, which halts model training and subsequent steps.

SUGGESTED_FIX:
Remove the `eval_metric="roc_auc"` argument from the `predictor.fit()` call. If you wish to specify the evaluation metric, use the `hyperparameters` dictionary (e.g., set `"optimization.metric_name": "roc_auc"`), or rely on AutoGluon's default metric selection for binary classification. Check the [AutoGluon MultiModal documentation](https://auto.gluon.ai/stable/tutorials/multimodal/index.html) for the correct way to set evaluation metrics.

Node 4 (debug)

Property	Value
uct_value	0.7739
avg_raw_score	0.6561666666666668
ctime	2026-05-22 17:28:58
debug_attempts	1
depth	2
execution_time	0.0
expected_child_count	0
exploitation	0.0555555555555556344
exploration	0.6065836461893294
failure_penalty	-0.05555555555555555
failure_visits	4
id	4
is_debug_successful	✗
is_leaf	✗
is_successful	✓
is_terminal	✗
normalized_failure_visit	1
normalized_score	0.16666666666666785
num_children	3
prev_tutorial_prompt	None
stage	debug
time_step	4
tool_used	machine learning
tools_available	['autogluon.multimodal', 'machine learning', 'autogluon.tabular']
unvalidated_visits	0
validated_contribution	0.1111111111111119
validated_reward	3.9370000000000007

validated_visits	6
validated_weight	0.6666666666666666
validation_score	0.7085
visits	10
parent_id	-1
child_ids	[8, 5, 6]

Node 5 (evolve)

Property	Value
uct_value	2.1456
ctime	2026-05-22 17:35:34
debug_attempts	0
depth	3
execution_time	0.0
expected_child_count	0
exploitation	0.0
exploration	1.482079962591042
failure_penalty	-0.0
failure_visits	1
id	5
is_debug_successful	✗
is_leaf	✓
is_successful	✗
is_terminal	✓
normalized_failure_visit	0
num_children	0
prev_tutorial_prompt	### Condensed: Intro to Use Machine Learning or Deep Learning algorithms # Condensed: Intro to Use Machine Learning or Deep Learning algorithms Summary: You can use any machine learning or deep learning algorithms to complete the tasks. Make sure yo # Intro to Use Machine Learning or Deep Learning algorithms You can use any machine learning or deep learning algorithms to complete the tasks. Make sure you have ther
stage	evolve
time_step	5
tool_used	machine learning

tools_available	['autogluon.multimodal', 'machine learning', 'autogluon.tabular']
unvalidated_visits	0
validated_contribution	0.0
validated_reward	0.0
validated_visits	0
validation_score	None
visits	1
parent_id	4
child_ids	[]

Error Message:

stderr:
==> WARNING: A newer version of conda exists. <==
current version: 25.11.0
latest version: 26.5.0
Please update conda by running
$ conda update -n base -c conda-forge conda
warning: Requirements file `/home/anri21/.conda/envs/mlzero_env/lib/python3.11/site-packages/autogluon/assistant/tools_registry/machine learning/requirements.txt` does not contain any dependencies
Using Python 3.11.15 environment at: 48-addition_2/node_5/conda_env
Resolved 79 packages in 356ms
Uninstalled 1 package in 1.91s
Installed 78 packages in 40.74s
+ aiohappyeyeballs==2.6.2
+ aiohttp==3.13.5
+ aiosignal==1.4.0
+ annotated-doc==0.0.4
+ annotated-types==0.7.0
+ anyio==4.13.0
+ attrs==26.1.0
+ certifi==2026.5.20
+ chardet==7.4.3
+ charset-normalizer==3.4.7
+ click==8.4.1
+ cuda-bindings==13.2.0
+ cuda-pathfinder==1.5.4
+ cuda-toolkit==13.0.2
+ datasets==4.8.5
+ dill==0.4.1
+ filelock==3.29.0

+ frozenlist==1.8.0
+ fsspec==2026.2.0
+ h11==0.16.0
+ hf-xet==1.5.0
+ httpcore==1.0.9
+ httpx==0.28.1
+ huggingface-hub==1.16.1
+ idna==3.16
+ jinja2==3.1.6
+ markdown-it-py==4.2.0
+ markupsafe==3.0.3
+ mdurl==0.1.2
+ mpmath==1.3.0
+ multidict==6.7.1
+ multiprocess==0.70.19
+ networkx==3.6.1
+ numpy==2.4.6
+ nvidia-cublas==13.1.1.3
+ nvidia-cuda-cupti==13.0.85
+ nvidia-cuda-nvrtc==13.0.88
+ nvidia-cuda-runtime==13.0.96
+ nvidia-cudnn-cu13==9.20.0.48
+ nvidia-cufft==12.0.0.61
+ nvidia-cufile==1.15.1.6
+ nvidia-curand==10.4.0.35
+ nvidia-cusolver==12.0.4.66
+ nvidia-cusparse==12.6.3.3
+ nvidia-cusparselt-cu13==0.8.1
+ nvidia-nccl-cu13==2.29.7
+ nvidia-nvjitlink==13.0.88
+ nvidia-nvshmem-cu13==3.4.5
+ nvidia-nvtx==13.0.85
+ opencv-python==4.13.0.92
+ pandas==3.0.3
+ pillow==12.2.0
+ propcache==0.5.2
+ pyarrow==24.0.0
+ pydantic==2.14.0a1
+ pydantic-core==2.47.0
+ pygments==2.20.0
+ python-calamine==0.6.2
+ python-dateutil==2.9.0.post0
+ pytz==2026.2
+ pyyaml==6.0.3
+ requests==2.34.2

+ rich==15.0.0
- setuptools==82.0.1
+ s... [truncated, 26941 chars total]

Error Analysis:

ERROR_SUMMARY: The error `joblib.externals.loky.process_executor.TerminatedWorkerError` indicates that one or more worker processes used for parallel computation were unexpectedly killed by the operating system. This is most commonly caused by excessive memory usage (e.g., flattening large images for all samples and using a high number of parallel jobs, such as n_jobs=64), which can lead to the OS terminating processes with SIGKILL (-9) to prevent system instability.

SUGGESTED_FIX:
Reduce the memory footprint by lowering the number of parallel jobs (set N_JOBS to a smaller value, such as 4 or 8), and/or decrease the image feature dimensionality (e.g., use a smaller resize shape like (32, 32) instead of (128, 128)). Rerun the script after these adjustments to avoid exhausting system memory and prevent worker termination.

Node 6 (evolve)

Property	Value
uct_value	1.0728
avg_raw_score	0.6457
ctime	2026-05-22 18:33:17
debug_attempts	0
depth	3
execution_time	0.0
expected_child_count	0
exploitation	0.0
exploration	1.2101132221170776
failure_penalty	-0.0
failure_visits	1
id	6
is_debug_successful	✗
is_leaf	✗
is_successful	✓
is_terminal	✗
normalized_failure_visit	0
normalized_score	0.0
num_children	2
prev_tutorial_prompt	### Condensed: Intro to Use Machine Learning or Deep Learning algorithms # Condensed: Intro to Use Machine Learning or Deep Learning algorithms Summary: You can use any machine learning or deep learning algorithms to complete the tasks. Make sure yo # Intro to Use Machine Learning or Deep Learning algorithms You can use any machine learning or deep learning algorithms to complete the tasks. Make sure you have ther
stage	evolve

time_step	6
tool_used	machine learning
tools_available	['autogluon.multimodal', 'machine learning', 'autogluon.tabular']
unvalidated_visits	0
validated_contribution	0.0
validated_reward	1.9371
validated_visits	3
validated_weight	1.0
validation_score	0.6457
visits	4
parent_id	4
child_ids	[16, 12]

Node 7 (debug)

Property	Value
uct_value	1.3384
ctime	2026-05-22 19:01:49
debug_attempts	3
depth	2
execution_time	0.0
expected_child_count	0
exploitation	0.0
exploration	1.664857771836881
failure_penalty	-0.0
failure_visits	2
id	7
is_debug_successful	✗
is_leaf	✗
is_successful	✗
is_terminal	✓
normalized_failure_visit	0
num_children	1
prev_tutorial_prompt	### Condensed: ```python # Condensed: ```python Summary: This tutorial demonstrates image classification using AutoGluon MultiModal, covering implementatio
stage	debug
time_step	7
tool_used	autogluon.multimodal
tools_available	['autogluon.multimodal', 'machine learning', 'autogluon.tabular']
unvalidated_visits	0
validated_contribution	0.0
validated_reward	0.0

validated_visits	0
validation_score	None
visits	2
parent_id	0
child_ids	[13]

Error Message:

stderr:
==> WARNING: A newer version of conda exists. <==
current version: 25.11.0
latest version: 26.5.0
Please update conda by running
$ conda update -n base -c conda-forge conda
Using Python 3.11.15 environment at: 48-addition_2/node_7/conda_env
Resolved 235 packages in 3.14s
Uninstalled 1 package in 2.18s
Installed 233 packages in 2m 14s
+ absl-py==2.4.0
+ accelerate==1.13.0
+ adagio==0.2.6
+ aiohappyeyeballs==2.6.2
+ aiohttp==3.13.5
+ aiohttp-cors==0.8.1
+ aiosignal==1.4.0
+ alembic==1.18.4
+ annotated-doc==0.0.4
+ annotated-types==0.7.0
+ antlr4-python3-runtime==4.9.3
+ attrs==26.1.0
+ autogluon==1.5.1b20260522
+ autogluon-common==1.5.1b20260522
+ autogluon-core==1.5.1b20260522
+ autogluon-features==1.5.1b20260522
+ autogluon-multimodal==1.5.1b20260522
+ autogluon-tabular==1.5.1b20260522
+ autogluon-timeseries==1.5.1b20260522
+ beartype==0.22.9
+ beautifulsoup4==4.14.3
+ blis==1.3.3
+ boto3==1.43.13
+ botocore==1.43.13
+ catalogue==2.0.10

+ catboost==1.2.10
+ certifi==2026.5.20
+ cffi==2.0.0
+ chardet==7.4.3
+ charset-normalizer==3.4.7
+ chronos-forecasting==2.2.2
+ click==8.4.1
+ cloudpathlib==0.24.0
+ cloudpickle==3.1.2
+ colorama==0.4.6
+ colorful==0.6.0a1
+ colorlog==6.10.1
+ confection==1.3.3
+ contourpy==1.3.3
+ coreforecast==0.0.16
+ cryptography==48.0.0
+ cycler==0.12.1
+ cymem==2.0.13
+ datasets==4.0.0
+ defusedxml==0.7.1
+ dill==0.3.8
+ distlib==0.4.0
+ einops==0.8.2
+ einx==0.4.3
+ evaluate==0.4.6
+ fastai==2.8.5
+ fastcore==1.8.18
+ fastdownload==0.0.7
+ fastprogress==1.0.5
+ fasttransform==0.0.2
+ filelock==3.29.0
+ fonttools==4.63.0
+ frozendict==2.4.7
+ frozenlist==1.8.0
+ fsspec==2025.3.0
+ fugue==0.9.7
+ future==1.0.0
+ gdown==6.0.0
+ gluonts==0.16.2
+ google-api-core==2.30.3
+ google-auth==2.53.0
+ googleapis-common-protos==1.75.0
+ graphviz==0.21
+ grpcio==1.81.0rc1
+ hf-xet==1.5.0

+ httpx==1.0.dev3
+ huggingface-hub==0.36.2
+ hyperopt==0.2.7
+ idna==3.16
... [truncated, 15817 chars total]

Error Analysis:

ERROR_SUMMARY: The error occurred during distributed training initialization with PyTorch Lightning's DDP strategy: torch.distributed.DistNetworkError ("address already in use" on port 15185). This means the TCP port required for inter-process communication is already occupied, likely due to a previous process not releasing the port or multiple jobs trying to use the same port simultaneously.

SUGGESTED_FIX:
Restart the node or ensure all previous training processes are terminated to free up the port. To avoid future conflicts, explicitly set a unique or random port for each run by adding the hyperparameter `"env.dist_addr": "localhost:"` to your predictor.fit() call, or set the environment variable `MASTER_PORT` to an unused port before running your script.

Node 8 (debug)

Property	Value
uct_value	1.0728
avg_raw_score	0.6457
ctime	2026-05-22 19:14:04
debug_attempts	1
depth	4
execution_time	0.0
expected_child_count	0
exploitation	0.0
exploration	1.0479887918088304
failure_penalty	-0.0
failure_visits	2
id	8
is_debug_successful	✗
is_leaf	✗
is_successful	✓
is_terminal	✗
normalized_failure_visit	0
normalized_score	0.0
num_children	2
prev_tutorial_prompt	### Condensed: Intro to Use Machine Learning or Deep Learning algorithms # Condensed: Intro to Use Machine Learning or Deep Learning algorithms Summary: You can use any machine learning or deep learning algorithms to complete the tasks. Make sure yo # Intro to Use Machine Learning or Deep Learning algorithms You can use any machine learning or deep learning algorithms to complete the tasks. Make sure you have ther
stage	debug

time_step	8
tool_used	machine learning
tools_available	['autogluon.multimodal', 'machine learning', 'autogluon.tabular']
unvalidated_visits	0
validated_contribution	0.0
validated_reward	1.2914
validated_visits	2
validated_weight	0.5
validation_score	0.6457
visits	4
parent_id	4
child_ids	[11, 14]

Node 9 (debug)

Property	Value
uct_value	1.4821
ctime	2026-05-22 19:42:27
debug_attempts	1
depth	2
execution_time	0.0
expected_child_count	0
failure_visits	1
id	9
is_debug_successful	✗
is_leaf	✓
is_successful	✗
is_terminal	✗
num_children	0
prev_tutorial_prompt	### Condensed: ```python # Condensed: ```python Summary: This tutorial demonstrates using AutoGluon Multimodal for multimodal classification on the PetFinde
stage	debug
time_step	9
tool_used	autogluon.tabular
tools_available	['autogluon.multimodal', 'machine learning', 'autogluon.tabular']
unvalidated_visits	0
validated_reward	0.0
validated_visits	0
validation_score	None
visits	1
parent_id	2
child_ids	[]

Error Message:

stderr:
==> WARNING: A newer version of conda exists. <==
current version: 25.11.0
latest version: 26.5.0
Please update conda by running
$ conda update -n base -c conda-forge conda
Using Python 3.11.15 environment at: 48-addition_2/node_9/conda_env
Resolved 239 packages in 1.29s
Uninstalled 1 package in 1.89s
Installed 237 packages in 1m 53s
+ absl-py==2.4.0
+ accelerate==1.13.0
+ adagio==0.2.6
+ aiohappyeyeballs==2.6.2
+ aiohttp==3.13.5
+ aiohttp-cors==0.8.1
+ aiosignal==1.4.0
+ alembic==1.18.4
+ annotated-doc==0.0.4
+ annotated-types==0.7.0
+ antlr4-python3-runtime==4.9.3
+ attrs==26.1.0
+ autogluon==1.5.1b20260522
+ autogluon-common==1.5.1b20260522
+ autogluon-core==1.5.1b20260522
+ autogluon-features==1.5.1b20260522
+ autogluon-multimodal==1.5.1b20260522
+ autogluon-tabular==1.5.1b20260522
+ autogluon-timeseries==1.5.1b20260522
+ beartype==0.22.9
+ beautifulsoup4==4.14.3
+ blis==1.3.3
+ boto3==1.43.13
+ botocore==1.43.13
+ catalogue==2.0.10
+ catboost==1.2.10
+ certifi==2026.5.20
+ cffi==2.0.0
+ chardet==7.4.3
+ charset-normalizer==3.4.7
+ chronos-forecasting==2.2.2
+ click==8.4.1
+ cloudpathlib==0.24.0

+ cloudpickle==3.1.2
+ colorama==0.4.6
+ colorful==0.6.0a1
+ colorlog==6.10.1
+ confection==1.3.3
+ contourpy==1.3.3
+ coreforecast==0.0.16
+ cryptography==48.0.0
+ cycler==0.12.1
+ cymem==2.0.13
+ daal==2025.9.0
+ datasets==4.0.0
+ defusedxml==0.7.1
+ dill==0.3.8
+ distlib==0.4.0
+ einops==0.8.2
+ einx==0.4.3
+ evaluate==0.4.6
+ fastai==2.8.5
+ fastcore==1.8.18
+ fastdownload==0.0.7
+ fastprogress==1.0.5
+ fasttransform==0.0.2
+ filelock==3.29.0
+ fonttools==4.63.0
+ frozendict==2.4.7
+ frozenlist==1.8.0
+ fsspec==2025.3.0
+ fugue==0.9.7
+ future==1.0.0
+ gdown==6.0.0
+ gluonts==0.16.2
+ google-api-core==2.30.3
+ google-auth==2.53.0
+ googleapis-common-protos==1.75.0
+ graphviz==0.21
+ grpcio==1.81.0rc1
+ hf-xet==1.5.0
+ httpx==1.0.dev3
+ huggingface-hub==0.36.2
+ hyperopt==0.... [truncated, 83478 chars total]

Error Analysis:

ERROR_SUMMARY: AutoGluon failed to train any models because Ray (used internally for parallelization) attempted to create UNIX domain socket files with paths exceeding the system limit of 107 bytes. This resulted in an OSError: "validate_socket_filename failed: AF_UNIX path length cannot exceed 107 bytes", causing all model training to abort and no predictions to be produced.

SUGGESTED_FIX:
Shorten the directory paths used for temporary files and output, especially the scratch and session directories (e.g., /sc-scratch/sc-scratch-ikim-guidlight/...). Move your working directory and all relevant data/output folders to a location with a much shorter absolute path (ideally under /tmp or a similarly short path), and update all script and environment variables accordingly. Then rerun the pipeline; this should allow Ray to create socket files with paths within the allowed length, resolving the OSError.

Node 10 (debug)

Property	Value
uct_value	1.4821
ctime	2026-05-22 19:51:13
debug_attempts	2
depth	3
execution_time	0.0
expected_child_count	0
failure_visits	1
id	10
is_debug_successful	✗
is_leaf	✓
is_successful	✗
is_terminal	✓
num_children	0
prev_tutorial_prompt	### Condensed: ```python # Condensed: ```python Summary: This tutorial demonstrates image classification using AutoGluon MultiModal, covering implementatio
stage	debug
time_step	10
tool_used	autogluon.multimodal
tools_available	['autogluon.multimodal', 'machine learning', 'autogluon.tabular']
unvalidated_visits	0
validated_reward	0.0
validated_visits	0
validation_score	None
visits	1
parent_id	3
child_ids	[]

Error Message:

stderr:
==> WARNING: A newer version of conda exists. <==
current version: 25.11.0
latest version: 26.5.0
Please update conda by running
$ conda update -n base -c conda-forge conda
Using Python 3.11.15 environment at: 48-addition_2/node_10/conda_env
Resolved 235 packages in 1.30s
Uninstalled 1 package in 1.93s
Installed 233 packages in 2m 06s
+ absl-py==2.4.0
+ accelerate==1.13.0
+ adagio==0.2.6
+ aiohappyeyeballs==2.6.2
+ aiohttp==3.13.5
+ aiohttp-cors==0.8.1
+ aiosignal==1.4.0
+ alembic==1.18.4
+ annotated-doc==0.0.4
+ annotated-types==0.7.0
+ antlr4-python3-runtime==4.9.3
+ attrs==26.1.0
+ autogluon==1.5.1b20260522
+ autogluon-common==1.5.1b20260522
+ autogluon-core==1.5.1b20260522
+ autogluon-features==1.5.1b20260522
+ autogluon-multimodal==1.5.1b20260522
+ autogluon-tabular==1.5.1b20260522
+ autogluon-timeseries==1.5.1b20260522
+ beartype==0.22.9
+ beautifulsoup4==4.14.3
+ blis==1.3.3
+ boto3==1.43.13
+ botocore==1.43.13
+ catalogue==2.0.10
+ catboost==1.2.10
+ certifi==2026.5.20
+ cffi==2.0.0
+ chardet==7.4.3
+ charset-normalizer==3.4.7
+ chronos-forecasting==2.2.2
+ click==8.4.1
+ cloudpathlib==0.24.0

+ cloudpickle==3.1.2
+ colorama==0.4.6
+ colorful==0.6.0a1
+ colorlog==6.10.1
+ confection==1.3.3
+ contourpy==1.3.3
+ coreforecast==0.0.16
+ cryptography==48.0.0
+ cycler==0.12.1
+ cymem==2.0.13
+ datasets==4.0.0
+ defusedxml==0.7.1
+ dill==0.3.8
+ distlib==0.4.0
+ einops==0.8.2
+ einx==0.4.3
+ evaluate==0.4.6
+ fastai==2.8.5
+ fastcore==1.8.18
+ fastdownload==0.0.7
+ fastprogress==1.0.5
+ fasttransform==0.0.2
+ filelock==3.29.0
+ fonttools==4.63.0
+ frozendict==2.4.7
+ frozenlist==1.8.0
+ fsspec==2025.3.0
+ fugue==0.9.7
+ future==1.0.0
+ gdown==6.0.0
+ gluonts==0.16.2
+ google-api-core==2.30.3
+ google-auth==2.53.0
+ googleapis-common-protos==1.75.0
+ graphviz==0.21
+ grpcio==1.81.0rc1
+ hf-xet==1.5.0
+ httpx==1.0.dev3
+ huggingface-hub==0.36.2
+ hyperopt==0.2.7
+ idna==3.16... [truncated, 12343 chars total]

Error Analysis:

ERROR_SUMMARY: The training process was terminated by a SIGTERM signal, which is typically sent when a job exceeds its allocated wall-clock time, hits resource limits (such as memory or GPU), or is manually killed by the system or scheduler. As a result, the model did not complete training, and no predictions or validation scores were generated.

SUGGESTED_FIX:
Check the total runtime of your script (including data loading, training, and inference) to ensure it stays well below the 1-hour wall-clock limit. Reduce training time by lowering the `time_limit` parameter in `predictor.fit()` (e.g., to 1200–1500 seconds), decreasing model complexity (e.g., use a smaller backbone or fewer epochs), or using fewer GPUs if resource contention is suspected. Also, monitor memory and GPU usage to ensure you are not exceeding hardware limits, and consider simplifying data preprocessing or batch sizes if necessary.

Node 11 (evolve)

Property	Value
uct_value	1.1772
ctime	2026-05-22 20:02:42
debug_attempts	1
depth	5
execution_time	0.0
expected_child_count	0
exploitation	0.0
exploration	1.482079962591042
failure_penalty	-0.0
failure_visits	2
id	11
is_debug_successful	✗
is_leaf	✗
is_successful	✗
is_terminal	✗
normalized_failure_visit	0
num_children	1
prev_tutorial_prompt	### Condensed: Intro to Use Machine Learning or Deep Learning algorithms # Condensed: Intro to Use Machine Learning or Deep Learning algorithms Summary: You can use any machine learning or deep learning algorithms to complete the tasks. Make sure yo # Intro to Use Machine Learning or Deep Learning algorithms You can use any machine learning or deep learning algorithms to complete the tasks. Make sure you have ther
stage	evolve
time_step	11
tool_used	machine learning

tools_available	['autogluon.multimodal', 'machine learning', 'autogluon.tabular']
unvalidated_visits	0
validated_contribution	0.0
validated_reward	0.0
validated_visits	0
validation_score	None
visits	2
parent_id	8
child_ids	[18]

Error Message:

stderr:
==> WARNING: A newer version of conda exists. <==
current version: 25.11.0
latest version: 26.5.0
Please update conda by running
$ conda update -n base -c conda-forge conda
warning: Requirements file `/home/anri21/.conda/envs/mlzero_env/lib/python3.11/site-packages/autogluon/assistant/tools_registry/machine learning/requirements.txt` does not contain any dependencies
Using Python 3.11.15 environment at: 48-addition_2/node_11/conda_env
Resolved 79 packages in 381ms
Uninstalled 1 package in 1.96s
Installed 78 packages in 42.03s
+ aiohappyeyeballs==2.6.2
+ aiohttp==3.13.5
+ aiosignal==1.4.0
+ annotated-doc==0.0.4
+ annotated-types==0.7.0
+ anyio==4.13.0
+ attrs==26.1.0
+ certifi==2026.5.20
+ chardet==7.4.3
+ charset-normalizer==3.4.7
+ click==8.4.1
+ cuda-bindings==13.2.0
+ cuda-pathfinder==1.5.4
+ cuda-toolkit==13.0.2
+ datasets==4.8.5
+ dill==0.4.1
+ filelock==3.29.0

+ frozenlist==1.8.0
+ fsspec==2026.2.0
+ h11==0.16.0
+ hf-xet==1.5.0
+ httpcore==1.0.9
+ httpx==0.28.1
+ huggingface-hub==1.16.1
+ idna==3.16
+ jinja2==3.1.6
+ markdown-it-py==4.2.0
+ markupsafe==3.0.3
+ mdurl==0.1.2
+ mpmath==1.3.0
+ multidict==6.7.1
+ multiprocess==0.70.19
+ networkx==3.6.1
+ numpy==2.4.6
+ nvidia-cublas==13.1.1.3
+ nvidia-cuda-cupti==13.0.85
+ nvidia-cuda-nvrtc==13.0.88
+ nvidia-cuda-runtime==13.0.96
+ nvidia-cudnn-cu13==9.20.0.48
+ nvidia-cufft==12.0.0.61
+ nvidia-cufile==1.15.1.6
+ nvidia-curand==10.4.0.35
+ nvidia-cusolver==12.0.4.66
+ nvidia-cusparse==12.6.3.3
+ nvidia-cusparselt-cu13==0.8.1
+ nvidia-nccl-cu13==2.29.7
+ nvidia-nvjitlink==13.0.88
+ nvidia-nvshmem-cu13==3.4.5
+ nvidia-nvtx==13.0.85
+ opencv-python==4.13.0.92
+ pandas==3.0.3
+ pillow==12.2.0
+ propcache==0.5.2
+ pyarrow==24.0.0
+ pydantic==2.14.0a1
+ pydantic-core==2.47.0
+ pygments==2.20.0
+ python-calamine==0.6.2
+ python-dateutil==2.9.0.post0
+ pytz==2026.2
+ pyyaml==6.0.3
+ requests==2.34.2

+ rich==15.0.0
- setuptools==82.0.1
+ ... [truncated, 2814 chars total]

Error Analysis:

ERROR_SUMMARY: The error occurred because the Python environment did not have the required package `autogluon` (specifically, `autogluon.multimodal`) installed, resulting in a `ModuleNotFoundError` when the script attempted to import it. The requirements files used during environment setup did not specify `autogluon`, so it was never installed.

SUGGESTED_FIX: Update your environment setup (bash script) to explicitly install `autogluon` (preferably `autogluon.multimodal`) using pip or conda after activating the environment and before running the Python script. Ensure the installation command (e.g., `pip install autogluon.multimodal`) is included regardless of the contents of the requirements files.

Node 12 (evolve)

Property	Value
uct_value	1.6649
avg_raw_score	0.6457
ctime	2026-05-22 20:07:17
debug_attempts	0
depth	4
execution_time	0.0
expected_child_count	0
exploitation	0.0
exploration	1.482079962591042
failure_penalty	-0.0
failure_visits	0
id	12
is_debug_successful	✗
is_leaf	✓
is_successful	✓
is_terminal	✗
normalized_failure_visit	0
normalized_score	0.0
num_children	0
prev_tutorial_prompt	### Condensed: Intro to Use Machine Learning or Deep Learning algorithms # Condensed: Intro to Use Machine Learning or Deep Learning algorithms Summary: You can use any machine learning or deep learning algorithms to complete the tasks. Make sure yo # Intro to Use Machine Learning or Deep Learning algorithms You can use any machine learning or deep learning algorithms to complete the tasks. Make sure you have ther
stage	evolve

time_step	12
tool_used	machine learning
tools_available	['autogluon.multimodal', 'machine learning', 'autogluon.tabular']
unvalidated_visits	0
validated_contribution	0.0
validated_reward	0.6457
validated_visits	1
validated_weight	1.0
validation_score	0.6457
visits	1
parent_id	6
child_ids	[]

Node 13 (debug)

Property	Value
uct_value	1.1772
ctime	2026-05-22 20:35:33
debug_attempts	3
depth	3
execution_time	0.0
expected_child_count	0
failure_visits	1
id	13
is_debug_successful	✗
is_leaf	✓
is_successful	✗
is_terminal	✓
num_children	0
prev_tutorial_prompt	### Condensed: ```python # Condensed: ```python Summary: This tutorial demonstrates image classification using AutoGluon MultiModal, covering implementatio
stage	debug
time_step	13
tool_used	autogluon.multimodal
tools_available	['autogluon.multimodal', 'machine learning', 'autogluon.tabular']
unvalidated_visits	0
validated_reward	0.0
validated_visits	0
validation_score	None
visits	1
parent_id	7
child_ids	[]

Error Message:

stderr:
==> WARNING: A newer version of conda exists. <==
current version: 25.11.0
latest version: 26.5.0
Please update conda by running
$ conda update -n base -c conda-forge conda
Using Python 3.11.15 environment at: 48-addition_2/node_13/conda_env
Resolved 235 packages in 4.23s
Uninstalled 1 package in 1.90s
Installed 233 packages in 2m 08s
+ absl-py==2.4.0
+ accelerate==1.13.0
+ adagio==0.2.6
+ aiohappyeyeballs==2.6.2
+ aiohttp==3.13.5
+ aiohttp-cors==0.8.1
+ aiosignal==1.4.0
+ alembic==1.18.4
+ annotated-doc==0.0.4
+ annotated-types==0.7.0
+ antlr4-python3-runtime==4.9.3
+ attrs==26.1.0
+ autogluon==1.5.1b20260522
+ autogluon-common==1.5.1b20260522
+ autogluon-core==1.5.1b20260522
+ autogluon-features==1.5.1b20260522
+ autogluon-multimodal==1.5.1b20260522
+ autogluon-tabular==1.5.1b20260522
+ autogluon-timeseries==1.5.1b20260522
+ beartype==0.22.9
+ beautifulsoup4==4.14.3
+ blis==1.3.3
+ boto3==1.43.13
+ botocore==1.43.13
+ catalogue==2.0.10
+ catboost==1.2.10
+ certifi==2026.5.20
+ cffi==2.0.0
+ chardet==7.4.3
+ charset-normalizer==3.4.7
+ chronos-forecasting==2.2.2
+ click==8.4.1
+ cloudpathlib==0.24.0

+ cloudpickle==3.1.2
+ colorama==0.4.6
+ colorful==0.6.0a1
+ colorlog==6.10.1
+ confection==1.3.3
+ contourpy==1.3.3
+ coreforecast==0.0.16
+ cryptography==48.0.0
+ cycler==0.12.1
+ cymem==2.0.13
+ datasets==4.0.0
+ defusedxml==0.7.1
+ dill==0.3.8
+ distlib==0.4.0
+ einops==0.8.2
+ einx==0.4.3
+ evaluate==0.4.6
+ fastai==2.8.5
+ fastcore==1.8.18
+ fastdownload==0.0.7
+ fastprogress==1.0.5
+ fasttransform==0.0.2
+ filelock==3.29.0
+ fonttools==4.63.0
+ frozendict==2.4.7
+ frozenlist==1.8.0
+ fsspec==2025.3.0
+ fugue==0.9.7
+ future==1.0.0
+ gdown==6.0.0
+ gluonts==0.16.2
+ google-api-core==2.30.3
+ google-auth==2.53.0
+ googleapis-common-protos==1.75.0
+ graphviz==0.21
+ grpcio==1.81.0rc1
+ hf-xet==1.5.0
+ httpx==1.0.dev3
+ huggingface-hub==0.36.2
+ hyperopt==0.2.7
+ idna==3.16... [truncated, 12542 chars total]

Error Analysis:

ERROR_SUMMARY: The error occurred because the training process was terminated by a SIGTERM signal, which was then escalated to a SIGKILL by the job scheduler (likely SLURM) due to exceeding resource constraints such as wall-clock time, memory, or GPU allocation. This happened during the `fit()` call of AutoGluon

MultiModalPredictor, preventing the model from completing training and producing predictions.

SUGGESTED_FIX:
Reduce the training time by lowering the `time_limit` parameter (e.g., set it to 900–1200 seconds), decrease model complexity (e.g., use a smaller backbone or fewer epochs), and/or reduce batch size or number of GPUs in the hyperparameters. Monitor resource usage and ensure the script (including data loading, training, and inference) completes well within the 1-hour wall-clock limit imposed by the scheduler.

Node 14 (evolve)

Property	Value
uct_value	1.6649
avg_raw_score	0.6457
ctime	2026-05-22 20:47:08
debug_attempts	0
depth	5
execution_time	0.0
expected_child_count	0
exploitation	0.0
exploration	1.482079962591042
failure_penalty	-0.0
failure_visits	0
id	14
is_debug_successful	✗
is_leaf	✓
is_successful	✓
is_terminal	✗
normalized_failure_visit	0
normalized_score	0.0
num_children	0
prev_tutorial_prompt	### Condensed: Intro to Use Machine Learning or Deep Learning algorithms # Condensed: Intro to Use Machine Learning or Deep Learning algorithms Summary: You can use any machine learning or deep learning algorithms to complete the tasks. Make sure yo # Intro to Use Machine Learning or Deep Learning algorithms You can use any machine learning or deep learning algorithms to complete the tasks. Make sure you have ther
stage	evolve

time_step	14
tool_used	machine learning
tools_available	['autogluon.multimodal', 'machine learning', 'autogluon.tabular']
unvalidated_visits	0
validated_contribution	0.0
validated_reward	0.6457
validated_visits	1
validated_weight	1.0
validation_score	0.6457
visits	1
parent_id	8
child_ids	[]

Node 15 (debug)

Property	Value
uct_value	1.4821
ctime	2026-05-22 21:14:43
debug_attempts	2
depth	2
execution_time	0.0
expected_child_count	0
failure_visits	1
id	15
is_debug_successful	✗
is_leaf	✓
is_successful	✗
is_terminal	✗
num_children	0
prev_tutorial_prompt	### Condensed: ```python # Condensed: ```python Summary: This tutorial demonstrates using AutoGluon Multimodal for multimodal classification on the PetFinde
stage	debug
time_step	15
tool_used	autogluon.tabular
tools_available	['autogluon.multimodal', 'machine learning', 'autogluon.tabular']
unvalidated_visits	0
validated_reward	0.0
validated_visits	0
validation_score	None
visits	1
parent_id	2
child_ids	[]

Error Message:

stderr:
==> WARNING: A newer version of conda exists. <==
current version: 25.11.0
latest version: 26.5.0
Please update conda by running
$ conda update -n base -c conda-forge conda
Using Python 3.11.15 environment at: 48-addition_2/node_15/conda_env
Resolved 239 packages in 3.14s
Uninstalled 1 package in 1.92s
Installed 237 packages in 1m 38s
+ absl-py==2.4.0
+ accelerate==1.13.0
+ adagio==0.2.6
+ aiohappyeyeballs==2.6.2
+ aiohttp==3.13.5
+ aiohttp-cors==0.8.1
+ aiosignal==1.4.0
+ alembic==1.18.4
+ annotated-doc==0.0.4
+ annotated-types==0.7.0
+ antlr4-python3-runtime==4.9.3
+ attrs==26.1.0
+ autogluon==1.5.1b20260522
+ autogluon-common==1.5.1b20260522
+ autogluon-core==1.5.1b20260522
+ autogluon-features==1.5.1b20260522
+ autogluon-multimodal==1.5.1b20260522
+ autogluon-tabular==1.5.1b20260522
+ autogluon-timeseries==1.5.1b20260522
+ beartype==0.22.9
+ beautifulsoup4==4.14.3
+ blis==1.3.3
+ boto3==1.43.13
+ botocore==1.43.13
+ catalogue==2.0.10
+ catboost==1.2.10
+ certifi==2026.5.20
+ cffi==2.0.0
+ chardet==7.4.3
+ charset-normalizer==3.4.7
+ chronos-forecasting==2.2.2
+ click==8.4.1
+ cloudpathlib==0.24.0

+ cloudpickle==3.1.2
+ colorama==0.4.6
+ colorful==0.6.0a1
+ colorlog==6.10.1
+ confection==1.3.3
+ contourpy==1.3.3
+ coreforecast==0.0.16
+ cryptography==48.0.0
+ cycler==0.12.1
+ cymem==2.0.13
+ daal==2025.9.0
+ datasets==4.0.0
+ defusedxml==0.7.1
+ dill==0.3.8
+ distlib==0.4.0
+ einops==0.8.2
+ einx==0.4.3
+ evaluate==0.4.6
+ fastai==2.8.5
+ fastcore==1.8.18
+ fastdownload==0.0.7
+ fastprogress==1.0.5
+ fasttransform==0.0.2
+ filelock==3.29.0
+ fonttools==4.63.0
+ frozendict==2.4.7
+ frozenlist==1.8.0
+ fsspec==2025.3.0
+ fugue==0.9.7
+ future==1.0.0
+ gdown==6.0.0
+ gluonts==0.16.2
+ google-api-core==2.30.3
+ google-auth==2.53.0
+ googleapis-common-protos==1.75.0
+ graphviz==0.21
+ grpcio==1.81.0rc1
+ hf-xet==1.5.0
+ httpx==1.0.dev3
+ huggingface-hub==0.36.2
+ hyperopt==0... [truncated, 83649 chars total]

Error Analysis:

ERROR_SUMMARY: AutoGluon failed to train any models because Ray attempted to create UNIX domain socket files with paths exceeding the 107-character limit imposed by the AF_UNIX standard. This occurred due to long temporary directory paths (e.g.,

`/sc-scratch/sc-scratch-ikim-guidlight/tmp/ray/session_.../sockets/plasma_store`), causing OSError and preventing model training.

SUGGESTED_FIX:
Set the `TMPDIR` (or `TMP`, `TEMP`) environment variable to a shorter path (e.g., `/tmp`) before running your script, so Ray and other libraries create temporary files in a directory with a shorter absolute path. Alternatively, move your working/output directories higher up in the filesystem hierarchy to ensure all generated socket paths remain under 107 characters.

Node 16 (evolve)

Property	Value
uct_value	1.1772
avg_raw_score	0.6457
ctime	2026-05-22 21:22:41
debug_attempts	0
depth	4
execution_time	0.0
expected_child_count	0
exploitation	0.0
exploration	1.482079962591042
failure_penalty	-0.0
failure_visits	1
id	16
is_debug_successful	✗
is_leaf	✗
is_successful	✓
is_terminal	✗
normalized_failure_visit	0
normalized_score	0.0
num_children	1
prev_tutorial_prompt	### Condensed: Intro to Use Machine Learning or Deep Learning algorithms # Condensed: Intro to Use Machine Learning or Deep Learning algorithms Summary: You can use any machine learning or deep learning algorithms to complete the tasks. Make sure yo # Intro to Use Machine Learning or Deep Learning algorithms You can use any machine learning or deep learning algorithms to complete the tasks. Make sure you have ther
stage	evolve

time_step	16
tool_used	machine learning
tools_available	['autogluon.multimodal', 'machine learning', 'autogluon.tabular']
unvalidated_visits	0
validated_contribution	0.0
validated_reward	0.6457
validated_visits	1
validated_weight	1.0
validation_score	0.6457
visits	2
parent_id	6
child_ids	[19]

Node 17 (debug)

Property	Value
uct_value	1.4821
ctime	2026-05-22 21:50:28
debug_attempts	3
depth	3
execution_time	0.0
expected_child_count	0
failure_visits	1
id	17
is_debug_successful	✗
is_leaf	✓
is_successful	✗
is_terminal	✓
num_children	0
prev_tutorial_prompt	### Condensed: ```python # Condensed: ```python Summary: This tutorial demonstrates image classification using AutoGluon MultiModal, covering implementatio
stage	debug
time_step	17
tool_used	autogluon.multimodal
tools_available	['autogluon.multimodal', 'machine learning', 'autogluon.tabular']
unvalidated_visits	0
validated_reward	0.0
validated_visits	0
validation_score	None
visits	1
parent_id	3
child_ids	[]

Error Message:

```
stderr:
==> WARNING: A newer version of conda exists. <==
current version: 25.11.0
latest version: 26.5.0
Please update conda by running
$ conda update -n base -c conda-forge conda
Using Python 3.11.15 environment at: 48-addition_2/node_17/conda_env
Resolved 235 packages in 1.32s
Uninstalled 1 package in 1.96s
Installed 233 packages in 2m 01s
+ absl-py==2.4.0
+ accelerate==1.13.0
+ adagio==0.2.6
+ aiohappyeyeballs==2.6.2
+ aiohttp==3.13.5
+ aiohttp-cors==0.8.1
+ aiosignal==1.4.0
+ alembic==1.18.4
+ annotated-doc==0.0.4
+ annotated-types==0.7.0
+ antlr4-python3-runtime==4.9.3
+ attrs==26.1.0
+ autogluon==1.5.1b20260522
+ autogluon-common==1.5.1b20260522
+ autogluon-core==1.5.1b20260522
+ autogluon-features==1.5.1b20260522
+ autogluon-multimodal==1.5.1b20260522
+ autogluon-tabular==1.5.1b20260522
+ autogluon-timeseries==1.5.1b20260522
+ beartype==0.22.9
+ beautifulsoup4==4.14.3
+ blis==1.3.3
+ boto3==1.43.14
+ botocore==1.43.14
+ catalogue==2.0.10
+ catboost==1.2.10
+ certifi==2026.5.20
+ cffi==2.0.0
+ chardet==7.4.3
+ charset-normalizer==3.4.7
+ chronos-forecasting==2.2.2
+ click==8.4.1
+ cloudpathlib==0.24.0
```

+ cloudpickle==3.1.2
+ colorama==0.4.6
+ colorful==0.6.0a1
+ colorlog==6.10.1
+ confection==1.3.3
+ contourpy==1.3.3
+ coreforecast==0.0.16
+ cryptography==48.0.0
+ cycler==0.12.1
+ cymem==2.0.13
+ datasets==4.0.0
+ defusedxml==0.7.1
+ dill==0.3.8
+ distlib==0.4.0
+ einops==0.8.2
+ einx==0.4.3
+ evaluate==0.4.6
+ fastai==2.8.5
+ fastcore==1.8.18
+ fastdownload==0.0.7
+ fastprogress==1.0.5
+ fasttransform==0.0.2
+ filelock==3.29.0
+ fonttools==4.63.0
+ frozendict==2.4.7
+ frozenlist==1.8.0
+ fsspec==2025.3.0
+ fugue==0.9.7
+ future==1.0.0
+ gdown==6.0.0
+ gluonts==0.16.2
+ google-api-core==2.30.3
+ google-auth==2.53.0
+ googleapis-common-protos==1.75.0
+ graphviz==0.21
+ grpcio==1.81.0rc1
+ hf-xet==1.5.0
+ httpx==1.0.dev3
+ huggingface-hub==0.36.2
+ hyperopt==0.2.7
+ idna==3.16... [truncated, 15755 chars total]

Error Analysis:

ERROR_SUMMARY: The error occurred because the distributed training backend (PyTorch DDP via Lightning) failed to initialize due to the requested TCP port (15185) already being in use (`EADDRINUSE`). This prevented the process group from being created, causing the entire training job to crash and halting all subsequent steps (prediction, validation, output file creation).

SUGGESTED_FIX:
Change the distributed training configuration to avoid port conflicts:
- Set the environment variable `MASTER_PORT` to a random or unused port before running your script, or
- Switch the `env.strategy` hyperparameter in AutoGluon to `"auto"` or `"ddp_spawn"` (or `"single"` if only one GPU is needed), or
- Limit training to a single GPU (`"env.num_gpus": 1`) to avoid distributed setup entirely.
After making these changes, rerun your script to ensure the port conflict is resolved.

Node 18 (debug)

Property	Value
uct_value	1.1772
ctime	2026-05-22 22:02:13
debug_attempts	1
depth	6
execution_time	0.0
expected_child_count	0
failure_visits	1
id	18
is_debug_successful	✗
is_leaf	✓
is_successful	✗
is_terminal	✗
num_children	0
prev_tutorial_prompt	### Condensed: Intro to Use Machine Learning or Deep Learning algorithms # Condensed: Intro to Use Machine Learning or Deep Learning algorithms Summary: You can use any machine learning or deep learning algorithms to complete the tasks. Make sure yo # Intro to Use Machine Learning or Deep Learning algorithms You can use any machine learning or deep learning algorithms to complete the tasks. Make sure you have ther
stage	debug
time_step	18
tool_used	machine learning
tools_available	['autogluon.multimodal', 'machine learning', 'autogluon.tabular']
unvalidated_visits	0
validated_reward	0.0
validated_visits	0

validation_score	None
visits	1
parent_id	11
child_ids	[]

Error Message:

stderr:
==> WARNING: A newer version of conda exists. <==
current version: 25.11.0
latest version: 26.5.0
Please update conda by running
$ conda update -n base -c conda-forge conda
warning: Requirements file `/home/anri21/.conda/envs/mlzero_env/lib/python3.11/site-packages/autogluon/assistant/tools_registry/machine learning/requirements.txt` does not contain any dependencies
Using Python 3.11.15 environment at: 48-addition_2/node_18/conda_env
Resolved 79 packages in 1.17s
Uninstalled 1 package in 1.80s
Installed 78 packages in 46.07s
+ aiohappyeyeballs==2.6.2
+ aiohttp==3.13.5
+ aiosignal==1.4.0
+ annotated-doc==0.0.4
+ annotated-types==0.7.0
+ anyio==4.13.0
+ attrs==26.1.0
+ certifi==2026.5.20
+ chardet==7.4.3
+ charset-normalizer==3.4.7
+ click==8.4.1
+ cuda-bindings==13.2.0
+ cuda-pathfinder==1.5.4
+ cuda-toolkit==13.0.2
+ datasets==4.8.5
+ dill==0.4.1
+ filelock==3.29.0
+ frozenlist==1.8.0
+ fsspec==2026.2.0
+ h11==0.16.0
+ hf-xet==1.5.0
+ httpcore==1.0.9
+ httpx==0.28.1
+ huggingface-hub==1.16.1
+ idna==3.16

+ jinja2==3.1.6
+ markdown-it-py==4.2.0
+ markupsafe==3.0.3
+ mdurl==0.1.2
+ mpmath==1.3.0
+ multidict==6.7.1
+ multiprocess==0.70.19
+ networkx==3.6.1
+ numpy==2.4.6
+ nvidia-cublas==13.1.1.3
+ nvidia-cuda-cupti==13.0.85
+ nvidia-cuda-nvrtc==13.0.88
+ nvidia-cuda-runtime==13.0.96
+ nvidia-cudnn-cu13==9.20.0.48
+ nvidia-cufft==12.0.0.61
+ nvidia-cufile==1.15.1.6
+ nvidia-curand==10.4.0.35
+ nvidia-cusolver==12.0.4.66
+ nvidia-cusparse==12.6.3.3
+ nvidia-cusparselt-cu13==0.8.1
+ nvidia-nccl-cu13==2.29.7
+ nvidia-nvjitlink==13.0.88
+ nvidia-nvshmem-cu13==3.4.5
+ nvidia-nvtx==13.0.85
+ opencv-python==4.13.0.92
+ pandas==3.0.3
+ pillow==12.2.0
+ propcache==0.5.2
+ pyarrow==24.0.0
+ pydantic==2.14.0a1
+ pydantic-core==2.47.0
+ pygments==2.20.0
+ python-calamine==0.6.2
+ python-dateutil==2.9.0.post0
+ pytz==2026.2
+ pyyaml==6.0.3
+ requests==2.34.2
+ rich==15.0.0
- setuptools==82.0.1
+ ... [truncated, 2781 chars total]

Error Analysis:

ERROR_SUMMARY: The error occurs because the Python environment does not have the `autogluon` package installed, resulting in `ModuleNotFoundError: No module named 'autogluon'` when the script attempts to import `MultiModalPredictor`. The requirements file used for installation does not specify `autogluon`, so it is missing from the environment.

SUGGESTED_FIX:
Update your bash script to explicitly install `autogluon` (e.g., by adding `pip install autogluon.multimodal` or `pip install autogluon` after installing other requirements). Ensure this installation step runs before executing the Python script so that all necessary modules are available.

Node 19 (evolve)

Property	Value
uct_value	1.1772
ctime	2026-05-22 22:06:14
debug_attempts	0
depth	5
execution_time	0.0
expected_child_count	0
failure_visits	1
id	19
is_debug_successful	✗
is_leaf	✓
is_successful	✗
is_terminal	✗
num_children	0
prev_tutorial_prompt	### Condensed: Intro to Use Machine Learning or Deep Learning algorithms # Condensed: Intro to Use Machine Learning or Deep Learning algorithms Summary: You can use any machine learning or deep learning algorithms to complete the tasks. Make sure you # Intro to Use Machine Learning or Deep Learning algorithms You can use any machine learning or deep learning algorithms to complete the tasks. Make sure you have ther
stage	evolve
time_step	19
tool_used	machine learning
tools_available	['autogluon.multimodal', 'machine learning', 'autogluon.tabular']
unvalidated_visits	0
validated_reward	0.0
validated_visits	0

validation_score	None
visits	1
parent_id	16
child_ids	[]

Error Message:

stderr:
==> WARNING: A newer version of conda exists. <==
current version: 25.11.0
latest version: 26.5.0
Please update conda by running
$ conda update -n base -c conda-forge conda
warning: Requirements file `/home/anri21/.conda/envs/mlzero_env/lib/python3.11/site-packages/autogluon/assistant/tools_registry/machine learning/requirements.txt` does not contain any dependencies
Using Python 3.11.15 environment at: 48-addition_2/node_19/conda_env
Resolved 79 packages in 354ms
Uninstalled 1 package in 1.78s
Installed 78 packages in 35.79s
+ aiohappyeyeballs==2.6.2
+ aiohttp==3.13.5
+ aiosignal==1.4.0
+ annotated-doc==0.0.4
+ annotated-types==0.7.0
+ anyio==4.13.0
+ attrs==26.1.0
+ certifi==2026.5.20
+ chardet==7.4.3
+ charset-normalizer==3.4.7
+ click==8.4.1
+ cuda-bindings==13.2.0
+ cuda-pathfinder==1.5.4
+ cuda-toolkit==13.0.2
+ datasets==4.8.5
+ dill==0.4.1
+ filelock==3.29.0
+ frozenlist==1.8.0
+ fsspec==2026.2.0
+ h11==0.16.0
+ hf-xet==1.5.0
+ httpcore==1.0.9
+ httpx==0.28.1
+ huggingface-hub==1.16.1
+ idna==3.16

+ jinja2==3.1.6
+ markdown-it-py==4.2.0
+ markupsafe==3.0.3
+ mdurl==0.1.2
+ mpmath==1.3.0
+ multidict==6.7.1
+ multiprocess==0.70.19
+ networkx==3.6.1
+ numpy==2.4.6
+ nvidia-cublas==13.1.1.3
+ nvidia-cuda-cupti==13.0.85
+ nvidia-cuda-nvrtc==13.0.88
+ nvidia-cuda-runtime==13.0.96
+ nvidia-cudnn-cu13==9.20.0.48
+ nvidia-cufft==12.0.0.61
+ nvidia-cufile==1.15.1.6
+ nvidia-curand==10.4.0.35
+ nvidia-cusolver==12.0.4.66
+ nvidia-cusparse==12.6.3.3
+ nvidia-cusparselt-cu13==0.8.1
+ nvidia-nccl-cu13==2.29.7
+ nvidia-nvjitlink==13.0.88
+ nvidia-nvshmem-cu13==3.4.5
+ nvidia-nvtx==13.0.85
+ opencv-python==4.13.0.92
+ pandas==3.0.3
+ pillow==12.2.0
+ propcache==0.5.2
+ pyarrow==24.0.0
+ pydantic==2.14.0a1
+ pydantic-core==2.47.0
+ pygments==2.20.0
+ python-calamine==0.6.2
+ python-dateutil==2.9.0.post0
+ pytz==2026.2
+ pyyaml==6.0.3
+ requests==2.34.2
+ rich==15.0.0
- setuptools==82.0.1
+ ... [truncated, 14328 chars total]

Error Analysis:

ERROR_SUMMARY: The error occurs because the test set provided to `predictor.predict_proba()` does not include all the columns that were present during training—specifically, the columns `['alternative_skin_tone', 'expert_opinion', 'skin_tone']` are missing from the test DataFrame. AutoGluon MultiModal requires that all columns used during training (even if only as metadata or tabular features) must also be present in the test set at prediction time, with placeholder values if necessary.

SUGGESTED_FIX:
Before calling `predictor.predict_proba(test_for_pred)`, add the missing columns (`'alternative_skin_tone'`, `'expert_opinion'`, `'skin_tone'`) to the test DataFrame, filling them with placeholder values (e.g., the mode or a default value such as `0` or `np.nan`). This ensures the test set schema matches the training set, allowing prediction to proceed without error.